team

SEUT

Satellite Edge Uplink Technology

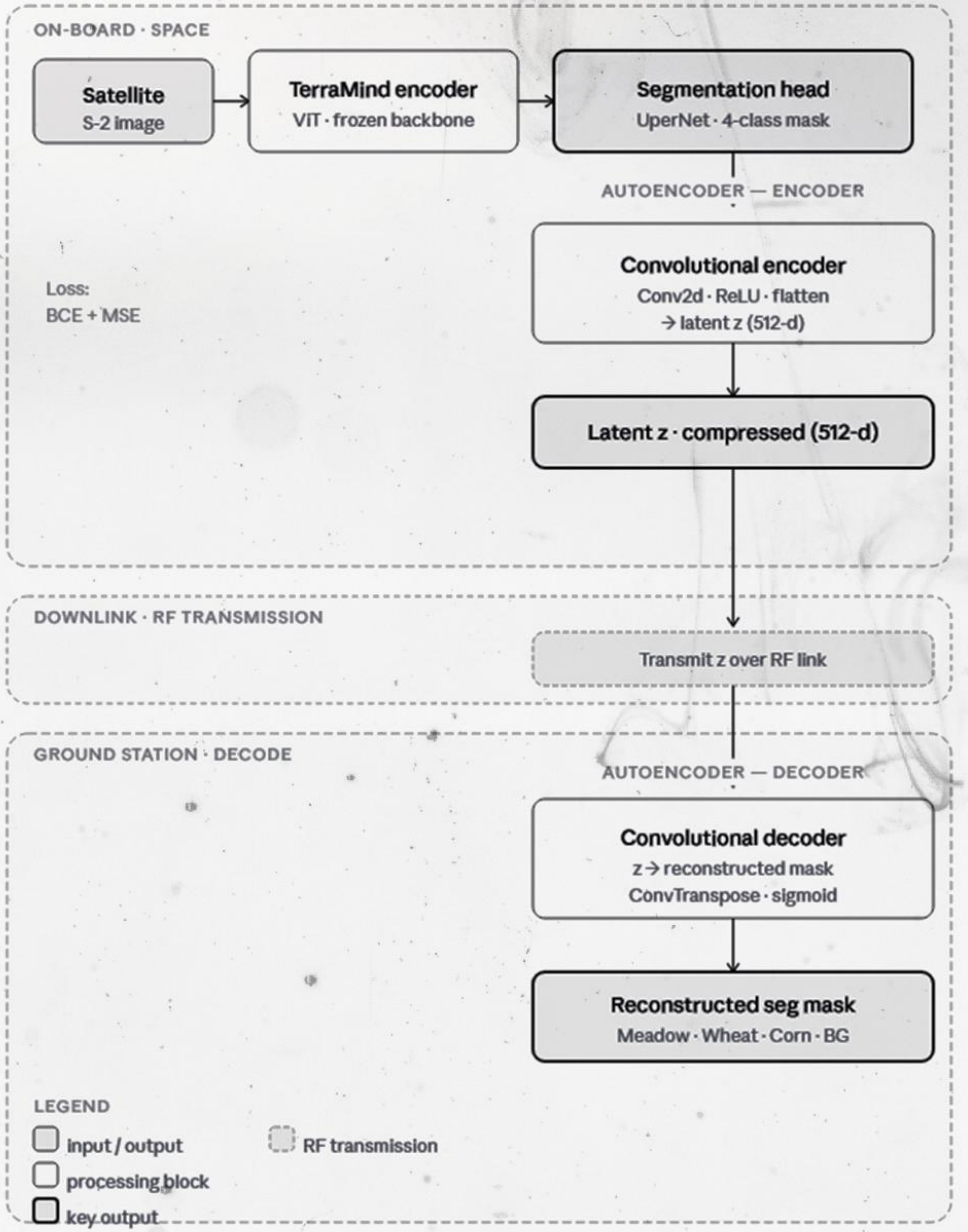
– T E R R A C R O P –

We've got a telescope in the heavens
but a straw for the data.
Transmitting the pulse of the planet
shouldn't cost the earth.

Dataset: PASTIS

Model: TerraMind + Segmentation Head

Fine-tuning method: Full fine-tuning – the entire model is completely unfrozen. TerraMind was updated at 1/10th the learning rate of the segmentation head (differential learning rates), preserving pretrained representations while still allowing the backbone to adapt.



TerraCrop | On-orbit crop intelligence · SEUT · TM2Space 2026

orbit 14,287 · 08:56:48 UTC · Link nominal

Raw mask	Compressed	Ratio	Fg accuracy	Seg mIoU
16,384 B	1536 B	10.7×	90.23%	0.54

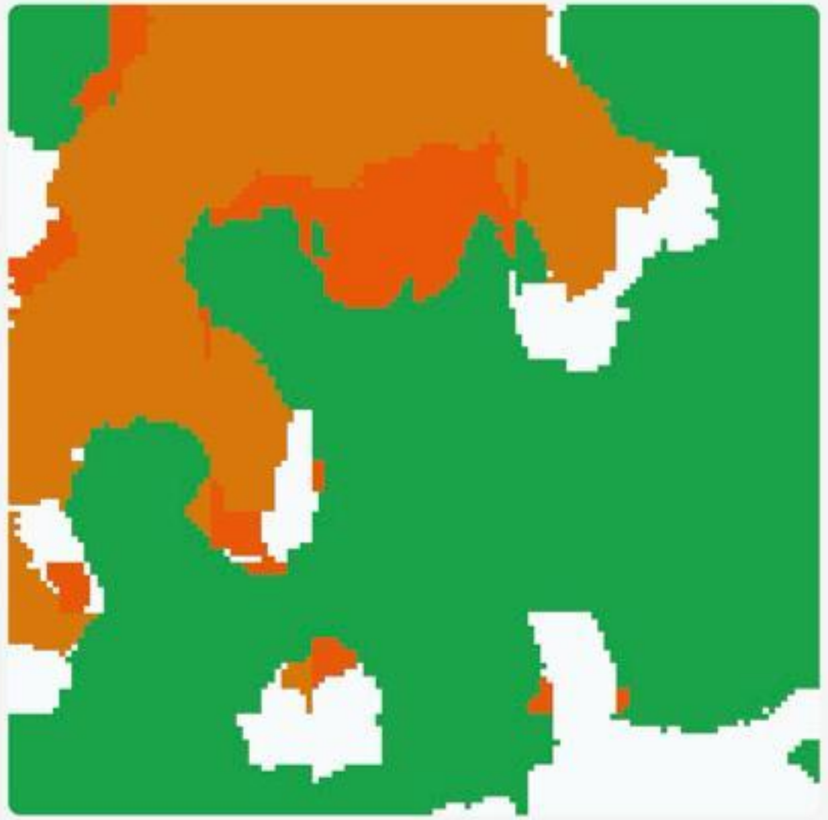
INPUT IMAGE

S2_40204.npy
12.5MB

S2_40204.npy
NIR · Red · Green false-colour composite

SEGMENTATION

Segmentation mask | Reconstructed mask



1536 bytes · 90.23% acc · 10.7× compression

Background	Meadow
Wheat	Corn

TELEMETRY

Raw mask	16,384 B
Compressed	1536 B
Ratio	10.7×
Px accuracy	79.91%
Fg accuracy	90.23%
Seg	123 ms
Encode	12 ms
Decode	2 ms

MODEL

Backbone	TerraMind-S
Dataset	PASTIS-HD
Latent dim	384d
Compressed	1536 B
Seg mIoU	0.54
Device	cuda

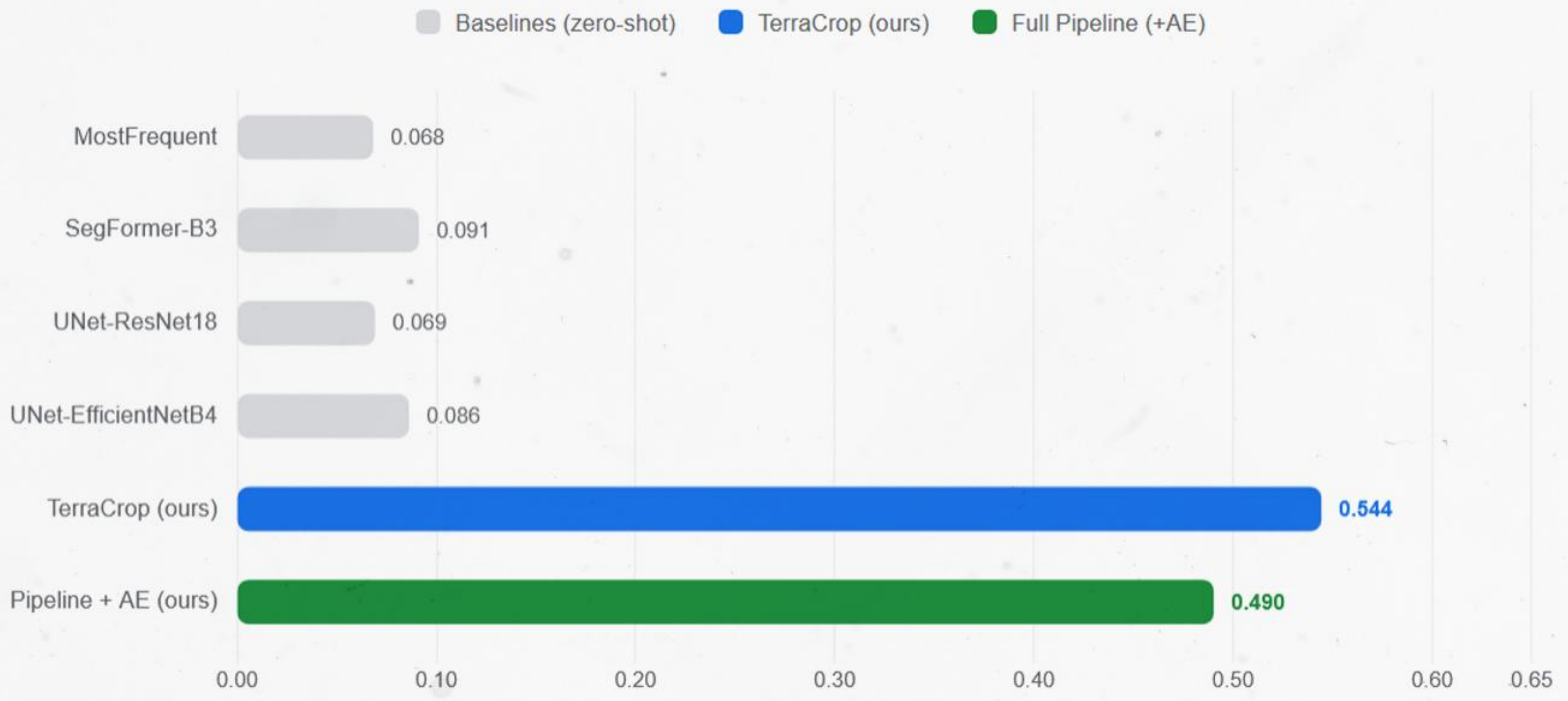
LOG

08:56:48	SUCCESS	Reconstructed · fg acc 90.23%
08:56:48	TRANSMIT	Payload 1536 B · S-band
08:56:48	COMPRESS	16,384 → 1536 B · 12 ms
08:56:48	INFER	S2_40204.npy · segmented 128×128
08:56:24	SYS	PASTIS-HD · 2,374 patches · 4 crop classes

TerraCrop Benchmark Results

MODEL	CATEGORY	PARAMS	MIOU	MEADOW	WHEAT	CORN	ACCURACY
MostFrequent	BASELINE	—	0.068	0.204	0.000	0.000	0.532
SegFormer-B3	BASELINE	44.6M	0.091	0.171	0.036	0.066	0.351
UNet-ResNet18	BASELINE	14.4M	0.069	0.132	0.076	0.000	0.303
UNet-EfficientNetB4	BASELINE	20.2M	0.086	0.116	0.062	0.081	0.243
TerraCrop (ours)	TERRAMIND	23.0M	**0.544**	**0.530**	**0.516**	**0.586**	0.855
Pipeline + AE (ours)	**FULL PIPELINE**	36.2M	**0.490**	**0.456**	**0.494**	**0.522**	**0.882**

AE compression loss = **0.054 mIoU** (0.544 → 0.490).



Key Insights

* 10×+ improvement over baselines in segmentation quality (mIoU from ~0.09 to 0.544)

* Compression introduces only ~5% loss (mIoU drops from 0.544 to 0.490)

* ~90% of semantic information retained after reconstruction

* Consistent performance across all crop classes

* Enables efficient transmission without losing critical agricultural insights

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